

Max Zhu

Atlanta, GA | mzhu351@gatech.edu | 404-940-3568 | maxzhu143.github.io

OBJECTIVE

Motivated, outgoing Computer Engineering student strongly interested in computer architecture, FPGA, and digital design. Seeking a Summer 2025 internship to gain real world experience in digital design, hardware-software integration, and low-level programming. Eager to contribute to innovative projects, determined to expand my knowledge, and excited to collaborate with a like-minded team to solve complex challenges in a dynamic environment.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

B.S. in Computer Engineering

Atlanta, GA

Expected Graduation May 2026

- **Threads:** Computing Hardware and Emerging Architecture and Systems and Architecture

GPA: 3.9/4.0

EXPERIENCE

Center for Research into Novel Computing Hierarchies Rogues Gallery

Reconfigurable Sub-Team Researcher

Atlanta, GA

January 2025 – Present

- Making a variant of the ChaCha20 program and implementing the cipher into an FPGA with Verilog

Georgia Tech School of Electrical and Computer Engineering

Opportunity Research Scholar for Wireless Twin Digital Systems

Atlanta, GA

August 2024 – Present

- Analyzed and annotated 50+ research papers on 5G, Wi-Fi 6, ray tracing, wireless signals, and UAVs
- Utilized a combination of Sionna, Blender, Wireless Insight, and radio environments create of wireless twin digital system

Metro Healthcare

IT Specialist

Vancouver, BC

August 2023 – January 2024

- Integrated digital infrastructure like POS systems, shared drives, website access, and patient database across multiple locations
- Developed an interactive UI with React.js to showcase clinical services and streamline appointment scheduling

PROJECTS

Cloud Server

July 2024 – Present

- Configured and deployed a cloud server with Ubuntu Server using only a CLI to replace multiple personal subscription services

Mbed Wordle Game

September 2024 – November 2024

- Created a Wordle-like game using C++, NAV Switch, Mbed, LPC1768, and an LCD for an interactive interface
- Used Google Tests to ensure correct back-end functionality and ensured no memory leaks

Spotify Car Thing

September 2024 – November 2024

- Developed a desktop device inspired by the “Spotify Car Thing” allowing for live interactions with the Spotify interface
- Integrated physical inputs and utilized the ESP8266, C++, and Spotify API to skip, like, pause, and view the current

SCOMP Processor and External Memory Device

August 2024 – November 2024

- Used VHDL to create a Simple Computer (SCOMP) processor and an external memory device using Model Sim to debug and a DE-10 Standard FPGA board to demonstrate correct functionality
- Implemented finite state machines, controllers and data paths in conjunction with each other

LEADERSHIP AND EXTRACURRICULARS

SILICON JACKETS

Digital Design Member

Atlanta, GA

August 2024 – Present

- Designed and implemented an arithmetic logic unit on a CPU using Verilog, focusing on state machines for optimized performance
- Developed with System Verilog and executed over 100+ tests to verify hardware designs, ensuring accuracy and reliability

HIVE

Peer Instructor

Atlanta, GA

December 2023 – Present

- Taught and guided end-users on laser cutters, 3d printers, shop machines, and benchtop tools
- Led interactive workshops on circuit design, skill implementation, and product design

SKILLS

Hardware: FPGA, Oscilloscopes, Digital Circuits, KiCad, Arduino, CAD, Software Defined Radio, Signal Analyzer

Software: C, C++, VHDL, Java, Python, Verilog, Git, CUDA, Ubuntu, MatLab, Sionna, Quartus, HTML, Javascript